INCH-POUND
MIL-DTL-24643/32E
22 August 2002
SUPERSEDING
MIL-C-24643/32D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS2SWAU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (watertight)

First -	Copper conductor, uncoated, ASTM B 286, AWG size No. 22, 7 strands.
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Crosslinked polyethylene insulation. Nominal wall thickness 0.013 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black and one white conductor for each pair.
Fourth -	Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form a pair.
Fifth -	Binder tape over each pair, applied at manufacturer's option, applied helically with overlap.
Sixth -	Braided shield of AWG No. 34 or No. 36 tin coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Seventh -	Shield insulation of two polyester tapes, per NEMA FI 4, outer tape sealed. An alternate shield insulation may consist of one polyester tape, per NEMA FI 4, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

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Construction (watertight) (continued)

Eighth - The required number of pairs (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used to make a firm, well rounded assembly.

Ninth - Binder tape, applied helically with overlap.

Tenth - Cross-linked polyolefin jacket (see table I for thickness).

TABLE I. Details.

Military part no. 1/ M24643/ 32	Size	Number of pairs	Cable jacket thickness min avg (inch)	Overall diameter (inches) TYPE LS2SWAU	
				min	max
-01UN	3	3	0.050	0.480	0.520
-02UN	7	7	0.050	0.610	0.660
-03UN	10	10	0.050	0.770	0.830
-04UN	14	14	0.063	0.860	0.930
-05UN	19	19	0.063	0.970	1.040
-06UN	24	24	0.063	1.120	1.210
-07UN	30	30	0.063	1.190	1.280
-08UN	37	37	0.063	1.290	1.380
-09UN	44	44	0.063	1.460	1.550
-10UN	61	61	0.075	1.660	1.740

1/ When double overall shield is required, see MIL-DTL-24643 for configuration of part number.

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum 17.37

Voltage withstand - volts, root mean square, minimum

Conductor to conductor 2000

Shield to shield (pair shields only) 500

Insulation resistance - Megohms -1000 feet minimum

Conductor to conductor 500

Component shield to shield 100

Conductor and shield continuity ..... No failure

Group A:

Visual and dimensional ..... No failure

Hydrostatic (open end) - leakage at 300 lb/in^2,

2 hours, in^3, maximum 1.0

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Capacitance
Mutual capacitance - 1 kilohertz (kHz), pF/ft, maximum 30
Capacitance unbalance - percent maximum 8
Characteristic impedance - at 1MHz, ohms..... 75 ± 5

Group B:
Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C).....50
Drip - 95 ± 1°C ..... Zero
Attenuation - at 3 MHz, decibels per 100 FT, maximum 3
Tear - pounds per inch thickness, minimum (ASTM D 470) 35
Physicals (unaged)
Insulation
Tensile strength - lb/in^2, minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Pulse response time - nanoseconds/500 FT, maximum 400
Physicals (aged)
Insulation
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125
Cable sealant removability..... No failure
Shield - conformance to material, construction and coverage..... No failure

Group D:
Flame propagation (cable) ..... No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Requirements:

Aging and compatibility (cable)(125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Size	Feet (nominal)
3 through 30	1000
37 through 61	500

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-030)

Review Activities:
Army - AR, AV, CR
Navy - CG, EC
DLA – CC

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